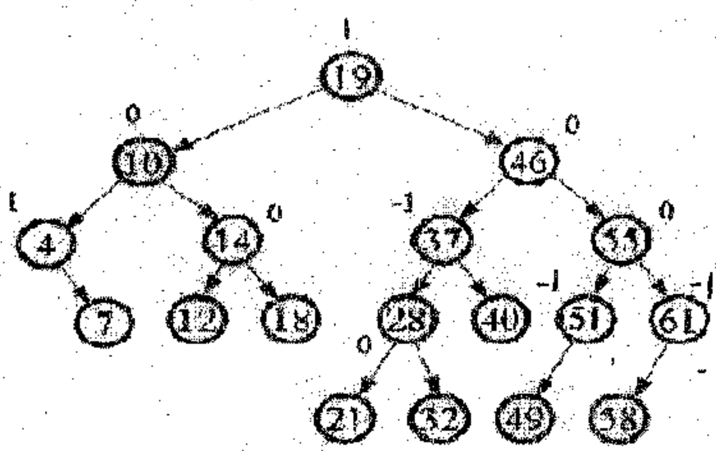
20.11.2017(E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
Nov – Dec 2017

Max. Marks: 100 | **Duration: 3hrs**
Class: SY | Semester: III
Name of the Course: Data Structures | Branch: COMP
Course Code: UCEC303

Instructions:
(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1	(a)For the given AVL tree Insert 34,2,11,9,56,20 And Delete 18,58,19,21,32  (Or) (a) Write a program to accept a graph and perform breadth first traversal of the graph. (b)Write a program to implement the Circular Linked List.	(10+10)
Q 2.	(a) Write a program to implement a priority queue using maximum heap. **(Or)** (a)Write an algorithm for Sparse matrix addition and construction. (b)Construct a B-tree of order 3 with the following set of elements. 45 29 32 49 63 18 27 30 31 36 39 46 47 54 59 61 67 72 Perform insertion and deletion with the following elements Insertion-1 5 7 11 13 15 17 19 Deletion-30 59 and 67	10+10

	(Or) (b)Write a program to implement Output Restricted Dequeue.	
Q 3.	(a)Write an algorithm for converting postfix expression to infix expression. And apply algorithm for the given expression. **a b + c - e f / * g h i / - -** (b)What is a Queue? Write an ADT for Queue. Explain the applications of Queue. **(Or)** (b)Write a Program to implement the polynomial addition and subtraction.	(10+10)
Q 4.	(a)Write a program to implement Selection Sort. (b)What is Hashing? Give its applications? Using quadratic probing insert the following values into the hash table of size 10. 28,35,71,67,11,10,90,44	(10+10)
Q 5.	(a) Write a program to read a given line of text, all the words starting with 'r' or 'R' should be reversed and the entire line is output. **(Sample Input: this is reverse program** **Sample Output: this is esrever program)** **(Or)** (a)Write a program to check whether the string has balanced parenthesis. Consider different parenthesis {, [, (, < 'in your program. (b) Write a program to implement construction of expression tree and its evaluation.	(10+10)

15.11.17 (E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)
End Semester Exam
NOV 2017

Max. Marks: 100 | **Duration: 3 Hrs**
Class: SYBTECH | Semester: III
Name of the Course: Applied Mathematics-III | Branch: COMP / IT
Course Code: UCEC301/ UITC301

Instructions:
(1) All Questions are Compulsory
(2) Figures to right indicate full marks.

<table>
<tr><th colspan="2">Question No.</th><th>Questions</th><th>Marks</th></tr>
<tr><td>Q.1</td><td>A</td><td>Find $L\{e^{-3t}\int_0^t t \sin 3t \ dt\}$</td><td>04</td></tr>
<tr><td>Q.1</td><td>B</td><td>Solve any THREE of the following.</td><td></td></tr>
<tr><td></td><td>(i)</td><td>a) Find L{t (2 sin 3t – 3 cos 3t)}
b) Find $L\{\frac{1-\cos t}{t}\}$</td><td>08</td></tr>
<tr><td></td><td>(ii)</td><td>a) Find $L^{-1}\left\{\frac{s+29}{(s+4)(s^2+9)}\right\}$
b) Find $L^{-1}\left\{\frac{2s^2-4}{(s+1)(s-2)(s-3)}\right\}$</td><td>08</td></tr>
<tr><td></td><td>(iii)</td><td>Solve by using Laplace transform
$\frac{dy}{dt}+2y+\int_0^t y(t)dt = \sin t$ with y(0)=1</td><td>08</td></tr>
<tr><td></td><td>(iv)</td><td>Using Convolution Theorem find $L^{-1}\left[\frac{(s+2)^2}{(s^2+4s+8)^2}\right]$</td><td>08</td></tr>
<tr><td></td><td>(v)</td><td>Prove that $\int_0^\infty e^{-\sqrt{2}\,t}\frac{\sin t \sin ht}{t}\,dt = \frac{\pi}{8}$</td><td>08</td></tr>
<tr><td>Q.2</td><td>A</td><td>Obtain the Fourier series of $f(x)=x(2\pi-x)$ in $0<x<2\pi$ of period 2π and hence deduce that $\sum_{n=1}^{\infty}\frac{1}{n^2}=\frac{\pi^2}{6}$</td><td>08</td></tr>
<tr><td>Q.2</td><td>B</td><td>Solve any TWO of the following.</td><td></td></tr>
<tr><td></td><td>(i)</td><td>Find Fourier series to represent f(x) $f(x)=4-x^2$ in the interval [0, 2].</td><td>08</td></tr>
<tr><td></td><td>(ii)</td><td></td><td>08</td></tr>
<tr><td></td><td>(iii)</td><td>Show that $x^2=\frac{\pi^2}{3}+4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos nx$ in $(-\pi,\pi)$ then deduce that $\sum_{n=1}^{\infty}\frac{1}{n^4}=\frac{\pi^4}{90}$</td><td>08</td></tr>
<tr><td>Q.3</td><td>A</td><td>Find eigen values & vectors of A^2 if A = $\begin{bmatrix}3 & 1 & -1\\ 2 & 2 & -1\\ 2 & 2 & 0\end{bmatrix}$</td><td>08</td></tr>
</table>

Q. 2 (B)
(ii) Find the Fourier cosine Transform of e^{-x}, $x \geq 0$ and hence deduce that $\int_0^\infty \frac{\cos mx}{1+x^2}dx = \frac{\pi}{2}e^{-m}$

(P.T.O.)

Q.3	**B**	**Solve any TWO of the following.**	
	(i)	State Cayley-Hamilton theorem & Verify it for the matrix A & hence find A^{-1} and A^4 if $A=\begin{bmatrix}1 & 2 & 3\\ 2 & -1 & 4\\ 3 & 1 & -1\end{bmatrix}$	**08**
	(ii)	Whether the matrix $A=\begin{bmatrix}8 & -8 & -2\\ 4 & -3 & -2\\ 3 & -4 & 1\end{bmatrix}$ is diagonalizable? If, yes find the diagonal matrix D and transforming matrix P.	**08**
	(iii)	a) Show that matrix $A=\begin{bmatrix}8 & -6 & 2\\ -6 & 7 & -4\\ 2 & -4 & 3\end{bmatrix}$ is non-derogatory. b) If $A=\begin{bmatrix}3 & 1\\ 1 & 3\end{bmatrix}$ find e^A.	**08**
Q.4	**A**	a) Prove that $\left[(\bar{a}\times\bar{b})\times(\bar{a}\times\bar{c})\right]\cdot\bar{d}=\left[\bar{a}\ \bar{b}\ \bar{c}\right](\bar{a}\cdot\bar{d})$ b) By considering the product $(\bar{a}\times\bar{b})\times(\bar{c}\times\bar{d})$ in two different ways, show that $\bar{d}=\frac{1}{[\bar{a}\ \bar{b}\ \bar{c}]}\left\{\left[\bar{b}\ \bar{c}\ \bar{d}\right]\bar{a}+\left[\bar{c}\ \bar{a}\ \bar{d}\right]\bar{b}+\left[\bar{a}\ \bar{b}\ \bar{d}\right]\bar{c}\right\}$ where $\bar{a},\bar{b},\bar{c}$ are non - coplaner	**08**
Q.4	**B**	**Solve any TWO of the following.**	
	(i)	a) Find $\nabla\cdot\bar{F}\ \&\ \nabla\times\bar{F}$ where $\bar{F}=\frac{x\hat{i}-y\hat{j}}{x^2+y^2}$ b) Prove that $\bar{F}=\frac{\bar{r}}{r^3}$ is both solenoidal and irrotational.	**08**
	(ii)	State Green's Theorem and using it evaluate $\int_C (y-sinx)dx+(cosx)dy$ where C is the triangle formed by the lines y=0, 2x=π and 2x= πy.	**08**
	(iii)	Evaluate the following integrals a) $\oint_S (ax\hat{\imath}+by\hat{\jmath}+cz\hat{k})\cdot d\bar{S}$ where S is $ax^2+bx^2+cz^2=abc$ (a, b,c are positive constants) b) $\int_S (\bar{\nabla}\times\bar{F})\cdot d\bar{S}$ where $\bar{F}=(x^2+y-4)\bar{\imath}+x\bar{\jmath}+(x+z^2)\bar{k}$, S is the hemisphere $x^2+y^2+z^2=a^2$, lying above xy-plane.	**08**

22-11-17 (E)

K. J. Somaiya College of Engineering, Mumbai-77
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End Semester Exam
Nov - Dec 2017

Max. Marks: 100 | **Duration: 3 hours**
Class: SY | Semester: III
Name of the Course: Computer Organization and Architecture | Branch: COMP
Course Code: UCEC304

Instructions:
(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1(a)	Draw the block diagram of Von Neumann model, explain in brief all the functional units.	10
Q 1(b)	Use booth's Algorithm for signed multiplication of following Multiplicand: +15 Multiplier: -6 Explain Booth's Algorithm with the help of above example.	10
Q2 (a)	What is the difference between restoring and non-restoring division algorithms? Demonstrate the following difference using given example.: 29 /3	10
Q2 (b)	Convert decimal representation to Single precision floating point representation(32 bit binary). (show stepwise solution) i. $(-307.1875)_{10}$ Also explain IEEE single precision standard format for floating point numbers with proper diagram.	10
Q3 (a)	Explain the instruction format and basic instruction cycle with the help of neat diagram. OR List the addressing modes and explain each with the help of an 8085 microprocessor instruction.	10
Q3 (b)	Explain Basic organization of a micro-programmed control unit with the help of block diagram and an example of a microinstruction routine. OR Write a note on Hardwired Control Unit and explain its advantages and disadvantages.	10
Q4 (a)	Write a note on cache architecture(L1,L2,L3) OR Write a note on MESI protocol	10

Q4 (b)	I) A 32 bit computer has a 32 bit memory address. It has 8Kb of cache memory. The computer follows four-way set associative mapping. Each line size is 16 bytes. Show the memory address format and cache memory organization. OR Find out page fault for following string using LRU and OPT 6,0,12,0,30,4,2,30,32,1,20,13 (consider page frame size = 3)	10
Q5 (a)	Write a note on following computer architectures i. SISD ii. SIMD iii. MISD iv. MIMD.	10
Q5 (b)	What is cache coherence; Explain any two approaches to maintain cache coherence in multiprocessor systems. OR Write a note on pipelining. Show that n stage pipe lined processor has n times speed up compared to non pipelined processor.	10

24-11-2017(E)

K. J. Somaiya College of Engineering, Mumbai-77
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End Semester Exam
Nov – Dec 2017

Max. Marks:100 **Duration:3 hrs**
Class: S.Y. Btech Semester: III
Name of the Course: Discrete Structure and Graph Theory Branch: COMPUTER
Course Code: UCEC305

Instructions:
(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Explain the following terms with suitable example (i) Disjoint Set (ii) Symmetric difference (iii)Partition Set (iv)Cartesian Product (v) Power Set	10
Q 1 (b)	List and explain any 5 laws of Set theory OR Explain with an example the Mutual Inclusion Exclusion Principle for three sets	10 10
Q2 (a)	Show that n3 + 2n is divisible by 3 for all n >=1	10
Q2 (b)	Define Universal and Existential Quantifiers with example. Transcribe the following into logical notation. Let the universe of discourse be real numbers (i) There are positive values of x and y such that x,y >0 (ii) For every value of x there is some value of y such that x . y =1 (iii)There is a value of x such that if y is positive then x+y is negative	10
Q3 (a)	Consider the set S= {1,2,3,4} and a relation R on S given by R= { (4,3), (2,2) , (2,1) , (3, 1) , (1, 2) } Show that R is not transitive and find transitive closure of R using Warshalls Algorithm OR Draw the Hasse diagram of D_{60} and check if it is a lattice	10 10

Q3 (b)	Define equivalence class. Let R be the following equivalence relation on the set A= {1,2,3,4,5,6} R= {(1,1), (1,5), (2,2,), (2,3), (2,6) ,(3,2) , (3,3), (3,6), (4,4), (5,1) , (5,5), (6,2) , (6,3), (6,6) } Find the partitions of A induced by R i.e find the equivalence classes of R	10
Q4 (a)	Define Pigeon hole principle. Show that if seven colors are used to paint 50 bicycles at least 8 cycles will be of the same color	10
Q4 (b)	Explain Eulerian and Hamiltonian path and circuit with suitable examples OR Define with an example (i) Planar Graph (ii) Isomorphic Graph	10 10
Q5 (a)	Consider (2,6) encoding function e: $B^2 \rightarrow B^6$ defined as e (00)= 000000 e (01)= 011110 e (10)=101010 e (11) =111000 (i) Find the minimum distance (ii) How many errors can ' e ' detect ? OR Define with an example : (i) Normal Subgroup (ii) Semigroup	10 10
Q5 (b)	Find the solution of recurrence relation $a_n = 5 a_{n-1} - 6 a_{n-2} + 7^n$	10

17-11-2017(E)

K. J. Somaiya College of Engineering, Mumbai-77

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End Semester Exam

Nov – Dec 2017

Max. Marks: | **Duration:**

Class: SY B.Tech | Semester: III

Name of the Course: Object Oriented Programming Methodology | Branch: Computer

Course Code: UCECE302

Instructions:

(1) All Questions are Compulsory

(2) Draw neat diagrams

(3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Explain Method overloading with example.	05
Q 1 (b)	Explain Cohesion and Coupling.	05
Q2 (a)	Define a class Point. Write a program that prompt user to input the x-y coordinate of point in Cartesian plane and using constructor initialize the values of x and y. The program should output a message indicating whether the point is the origin or is located on the X (or Y) axis, or appears in a particular quadrant using tostring function. For Example: (0, 0) is the Origin, (4, 0) is on the X-Axis, (-2, 3) is in second quadrant. **OR** Write a program to store shopping items in vector. Accept 5 items from command prompt. And store it in vector and also provide following functions: 1. Add the item in the vector 2. Remove item from the vector 3. Display Vector using Enumeration	10 10
Q.2(b)	A production company maintains the record of all products manufactured in the company. The product information like product ID, name, date of manufacturing etc. Define a class for the product. And make use of array of objects and write a functions to perform following operations: 1. Store new product information in an array of objects 2. Display all product details 3. Delete the product from array. **OR**	10

Q.2 (b) (10 marks)

Write a program for Tic-Tac-Toe game. Consider 3by 3 Matrix and accept the position from user for one player and one player is computer. For computer move make use of Random class. For all users display the output after every input(Position) and display the winner.(Give the numbers to each block and accept the block number from user for X and 0). Following table shows Block structure:

1	2	3
4	5	6
7	8	9

Q3 (a) (10 marks)

Find the longest common subsequence of two strings using following algorithm accept two strings from user and display longest common subsequence.

S(j,k)=length of common subsequence of first j character of X and first k character of Y.

S is 2D array. S(X.length(), Y.length())

S(0,k)=s(j,0)=0

If j^{th} character of X matches with k^{th} character of Y then

S(j,k)=s(j-1, k-1)+1

Make the j^{th} character of X and k^{th} character of Y mark it bold

If j^{th} character of X do not match with k^{th} character of Y then

S(j,k)=the larger of s(j,k-1) and s(j-1,k)

For Example:

X=GTTCG Y=AGCTACCG

X/ Y	-	A	**G**	C	**T**	A	C	**C**	**G**
-	0	**0**	0	0	0	0	0	0	0
G	0	0	**1**	1	1	1	1	1	1
T	0	0	1	**1**	2	2	2	2	2
T	0	0	1	1	**2**	**2**	**2**	2	2
C	0	0	1	1	2	2	3	**3**	3
G	0	0	1	1	2	2	3	3	**4**

Longest Common subsequence of X and Y is **G-T-C-G**

Q.3 (b) (10 marks)

Explain ArrayList and Wrapper class with example.

Q4 (a)	Draw the class diagram for college. It consists of following classes: college, student, department, lab, and teacher, staff. Identify the attributes and functions of each class. Assume suitable data (if required). Write your assumptions clearly. Show the relations and label the relations properly. Also show the multiplicity according to your assumption.	10
Q4 (b)	Explain User defined Packages and how packages are used in visibility control with proper example.	10
Q5 (a)	Write a static function to accept an integer number from user. If a number consist of digit 9 then the exception should be thrown but not caught in function (i.e. no explicit catch block). Write a client program that calls this method and print "Error: Number having digit 9" if the method throws exception. **OR** Explain the Exception handling in Java	10
Q5 (b)	Explain Thread life cycle. Write a program to display characters [A-Z] and display of digits [0-9]. Create one thread for displaying characters and other thread to display the digits. Make sure that the main thread ends after display of all characters and numbers.	10
Q5 (c)	Explain the life cycle of Applet. Write a code to display smiley face. **OR** Display Digital Clock using Applet programming in java. Make use of paint and repaint method of applet for continuously changing the time. Display time in HH:MM:SS format.	10